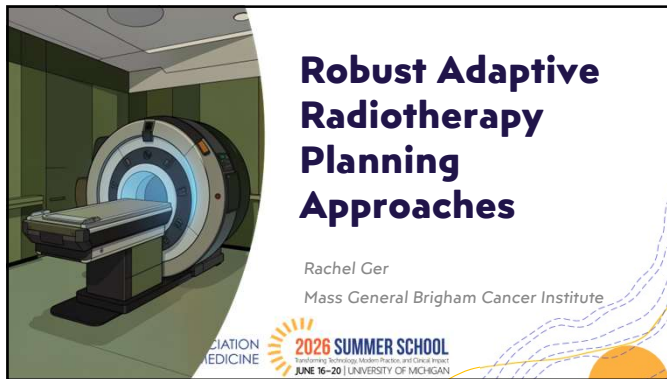
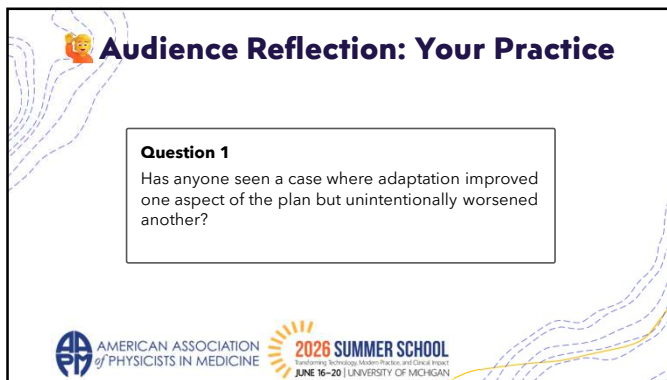




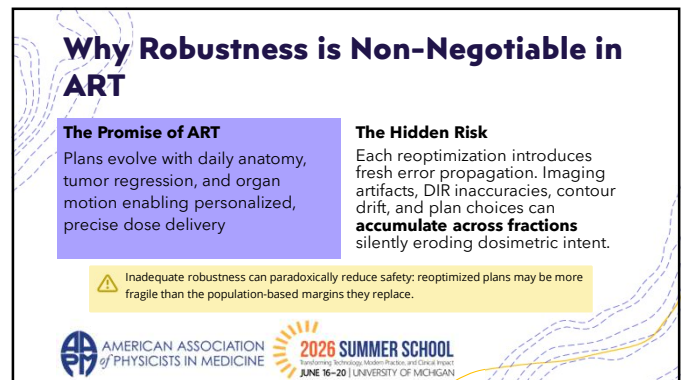
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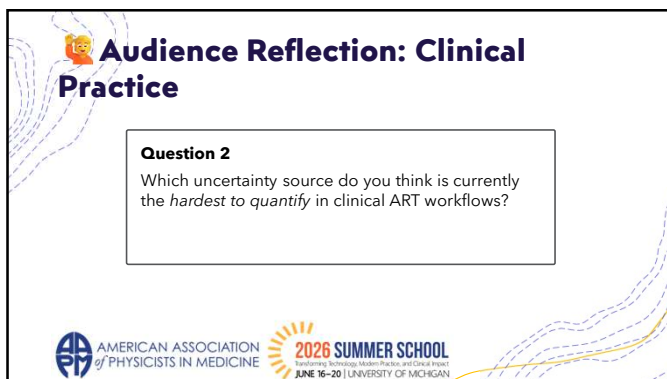
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Sources of Uncertainty in ART

Imaging & Contouring CBCT HU instability, DIR assumptions violated in real anatomy, deep-learning reconstruction bias	Anatomical Changes Tumor regression, weight loss, edema, parotid migration, bladder/rectal filling variation
Setup & Motion Residual systematic and random errors; respiratory interplay in thoracic/abdominal sites	Proton-Specific Range uncertainty from HU-to-stopping-power conversion; LET variation near Bragg peak; anatomical density shifts

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Robust Optimization Strategies

Method	Pros	Cons	Best fit	Decision cue
Worst-case	Maximum safety	Plan quality loss	High-risk anatomy, tight OAR constraints	Choose when safety margin matters most.
Probabilistic	Balanced	Computational	Well-characterized setup/range uncertainty	Choose when you want balance with quantifiable risk.
Multi-scenario	Clinically intuitive	Manual effort	Workflow-friendly scenario-based planning	Choose when adaptation must be practical at the console.
CVaR	Tail-risk focused	Complex	When worst-case outliers drive concern	Choose when you want to control the worst tail without over-penalizing all scenarios.

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Audience Reflection: Discussion

Question 3

Take 60 seconds and discuss with the person next to you: should robustness be prioritized over conformity in adaptive planning?

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Specific Implementations: Protons

- Utilization of CBCT requires synthetic CT generation
- Defining setup and range uncertainty scenarios (typically $\pm 3-5\text{mm}$ and $\pm 3-3.5\%$)
- QA/QC workflow can mitigate risks of sCT generation and/or be used to verify plan generated on sCT

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Specific Implementations: MRI

- DIR utilization from planning CT/MR to daily MR must be robust
- Magnetic field effects on dose
- Intrafraction motion can degrade plan's effectiveness

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Where the Field is Heading

- **AI-Driven Automation**
Predictive anatomical models, auto-contouring, and reinforcement learning for near-real-time robust replanning
- **Biological Robustness**
NTCP-guided plan selection, variable RBE/LET optimization, voxel-level dose painting integrated into robust frameworks
- **Imaging-Driven Uncertainty**
sCT-specific range uncertainty terms, ensemble sCT generation, and voxel-level HU uncertainty maps folded into scenario generation
- **Standardization**
Consensus robust DVH metrics, multi-institutional trials, and professional society task groups to harmonize robustness thresholds across centers

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Audience Reflection: Future Clinical Practice

Question 5

What advancement will most transform robust ART over the next 5-10 years?

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Audience Reflection: Future Clinical Practice

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Core Takeaways

- 🕒 **Robustness is a prerequisite** — not an optional safeguard — for realizing the promise of ART. Dynamic reoptimized plans can be more fragile than static margins if uncertainty is not explicitly modeled.
- 🕒 Each ART platform demands **platform-specific robust implementation**: sCT-based stopping power for protons, electron return effect correction for MR-linacs, contour uncertainty quantification for all.
- 🕒 The **ultimate vision**: a seamless intelligent adaptive workflow where robustness is continuously monitored, updated, and integrated into every treatment decision — precision and resilience unified.

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